

AMD-ITRI HIGH-PERFORMANCE COMPUTING PLATFORM

MI300X RAG Project Report

Enterprise document retrieval and source-grounded Chinese answer generation

AMD
INSTINCT™
MI300X

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A complete RAG pipeline

Data preparation

- PDF text extraction using pypdf
- Excel reading using openpyxl
- 500-character chunks with 100-character overlap

Retrieval model

BAAI/bge-small-zh-v1.5

- Chinese semantic embeddings
- Normalized vectors and Top-3 retrieval

Generation model

Qwen2.5-7B-Instruct

- bfloat16 inference on MI300X
- Traditional Chinese answer with sources

The project makes enterprise documents searchable and answerable

Problem addressed

- Information is distributed across PDF and Excel files
- Manual information finding is time-consuming
- Answers need traceable document sources

Implemented objective

- Build a searchable Chinese document knowledge index
- Retrieve relevant document chunks for a question
- Generate a source-grounded answer with Qwen

Document content becomes a persistent semantic index

1. Read documents

PDF pages
Excel rows and sheets

2. Create chunks

Clean text
500-character chunks
100-character overlap

3. Save index

gama_embeddings.npy
gama_chunks.json
index_summary.json

Verified index output



Reachable chunks.

The architecture links document evidence to the generated answer



The two models have distinct roles in the RAG pipeline

Embedding & Retrieval

BAAI/bge-small-zh-v1.5

- Transforms document chunks and the user question into comparable vectors
- Normalized vectors; Top-3 by dot-product similarity

Answer generation

Qwen2.5-7B-Instruct

- Receives retrieved context and system instruction
- Produces Traditional Chinese answer and source filenames

MI300X results and comparison

25

document sections
extracted

17

searchable text chunks
created

2.55 s

observed Qwen generation
time

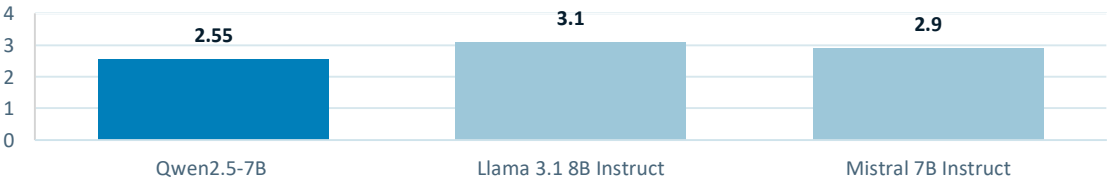
Top 3

retrieved chunks per query

Execution environment

AMD Instinct MI300X • PyTorch 2.3 • ROCm 6.2

Response time comparison (seconds)



Proposed deployment model

Final product functions

- Controlled PDF / Excel upload
- Document versioning and indexing
- Source-grounded Q&A
- Source citations and user feedback



NEXT STEP

Request for renewed MI300X access

Planned work with renewed access

- Expand official product and service documents
- Create a 50–100 question evaluation set
- Measure retrieval and inference performance systematically

Expected deliverables

- Reproducible MI300X benchmark report
- Validated source-grounded RAG results
- A proposed interface for enterprise knowledge support